

Augmented Lagrangian Predictive Coding

From local prediction errors to backpropagation credit

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sPC

Relax hidden states

ePC

Optimize error coordinates

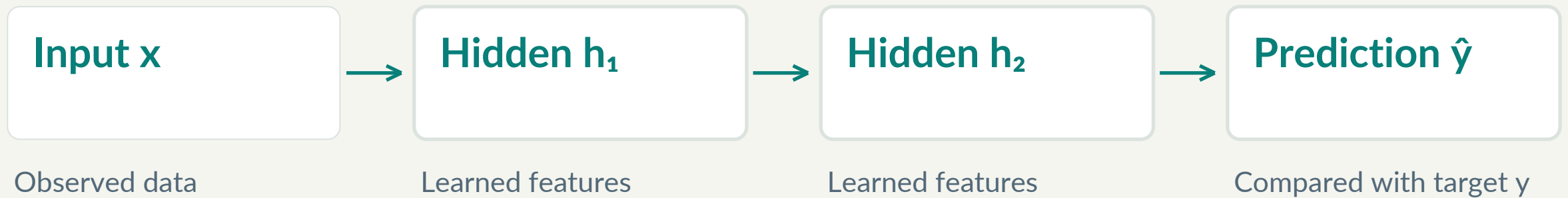
PC-ALM

Accumulate local dual credit

30-MINUTE TEAM PRESENTATION · 27 MIN TALK + 3 MIN DISCUSSION

First: what are we trying to learn?

A forward computation and a loss define the learning problem.



$$h_i = \sigma(W_i h_{i-1}), \quad \hat{y} = W_L h_{L-1}$$

$$\ell = \frac{1}{2} \|y - \hat{y}\|^2$$

Weights W are shared. Hidden states h belong to an example.

Predictive coding makes hidden activations adjustable during training.

Backpropagation assigns credit by the chain rule

Each hidden layer receives the derivative of the downstream loss.

The derivative we need

How much would the final loss change if this hidden activation changed?

δ_i : BP derivative of loss with respect to h_i

$$\delta_{L-1} = W_L^T(\hat{y} - y)$$

$$\delta_i = J_{i+1}^T \delta_{i+1}$$

J_{i+1} is the adjacent layer's Jacobian: its table of local sensitivities.

BP computes these signals in a reverse pass through the composed network.

Standard PC lets hidden states move

Inference minimizes a supervised loss plus layerwise prediction errors.

$$r_i = h_i - \sigma(W_i h_{i-1})$$

r = state – prediction $\rho > 0$ = penalty strength T = inference steps

$$F_{\text{PC}} = \ell + \frac{\rho}{2} \sum_{i=1}^{L-1} \|r_i\|^2$$

1 Infer states

Fixed weights; take T steps on h .

2 Learn weights

Fixed final states; use local gradients.

An interior activity gradient depends only on its own and adjacent layer terms.

A finite penalty changes the equilibrium

The hidden states can improve the loss by violating the forward equations.

| A compromise is allowed

Lower the supervised loss by moving a hidden state away from its forward prediction.

ONE HIDDEN SCALAR

$$F_{\text{PC}}(h) = \frac{1}{2}(1 - h)^2 + \frac{1}{2}(h - 0.2)^2$$

Forward hidden state **0.2**

Settled PC hidden state **0.6**

Faster inference alone cannot remove a finite-penalty difference from BP.

ePC optimizes prediction errors directly

Same energy on an acyclic graph; a different route through it.

$$h_i(\varepsilon) = \sigma(W_i h_{i-1}(\varepsilon)) + \varepsilon_i$$

$$E(\varepsilon) = F_{\text{PC}}(h(\varepsilon)), \quad \varepsilon \leftarrow \varepsilon - \eta_{\varepsilon} \nabla_{\varepsilon} E$$

Forward: errors $\rightarrow$ states

An early error changes downstream states.

Reverse: energy $\rightarrow$ error gradients

FabricPC differentiates the full derived graph.

ePC uses global error inference and retains a local final weight update.

ePC can be BP-like before it equilibrates

A quick learning signal and a converged PC solution are different endpoints.

| Small error step

From zero hidden errors, the first error gradient is the BP activation derivative.

Weight gradients are BP-like to first order.

| Near PC equilibrium

More relaxation solves the PC energy's tradeoff.

Stable convergence does not make that objective equal to BP's.

Compare both ePC regimes; report step size, optimizer, and curvature stability.

Add a multiplier that remembers violations

The augmented Lagrangian combines a task loss, constraints, and a penalty.

$$\mathcal{L}_\rho = \ell + \sum_i \lambda_i^\top r_i + \frac{\rho}{2} \sum_i \|r_i\|^2$$

Task loss ℓ

Fits the observed target.

Multiplier λ

Accumulates each hidden constraint's violations.

Penalty ρ

Supplies quadratic curvature.

Descend in hidden states h ; ascend in sample-specific multipliers λ .

One batch: relax, accumulate, then learn

Algorithm 1 uses T activity steps and $T-1$ multiplier updates.

- 01 Initialize** Forward states; hidden multipliers $\lambda = 0$.
- 02 Primal step** All free states descend the augmented objective.
- 03 Dual step** Recompute new-state residuals; $\lambda \leftarrow \lambda + \alpha r$.
- 04 Finish and learn** Final primal only; recompute local weight gradients.

Repeat steps 02–03 exactly $T-1$ times. Weights stay fixed until 04.

The dual step reads NEW residuals. α is a separate rate; each new batch resets λ .

Replace the residual credit with a composite

Both activity inference and hidden-weight learning use $c = \lambda + pr$.

COMPOSITE CREDIT

$$c_i = \lambda_i + pr_i$$

λ = accumulated violation
 pr = current mismatch pressure

$$\nabla_{h_i} \mathcal{L}_\rho = c_i - J_{i+1}^\top c_{i+1}$$

$$\nabla_{W_i} \mathcal{L}_\rho = -(c_i \odot \sigma'(W_i h_{i-1})) h_{i-1}^\top$$

Use the augmented signal in BOTH inference and learning. Keep the true residual separate.

Why the endpoint gives the BP gradient

Zero residuals restore forward states; nonzero multipliers retain credit.

1 Feasibility

$$r_i = 0$$

States return to forward values.

2 Stationarity

$$\lambda_i = -\delta_i$$

Multipliers reproduce BP credit.

3 Learning

$$\nabla_{\theta} \mathcal{L}_{\rho} = \nabla_{\theta} \ell$$

Local weight gradients equal BP.

Endpoint identity: differentiable feedforward maps.

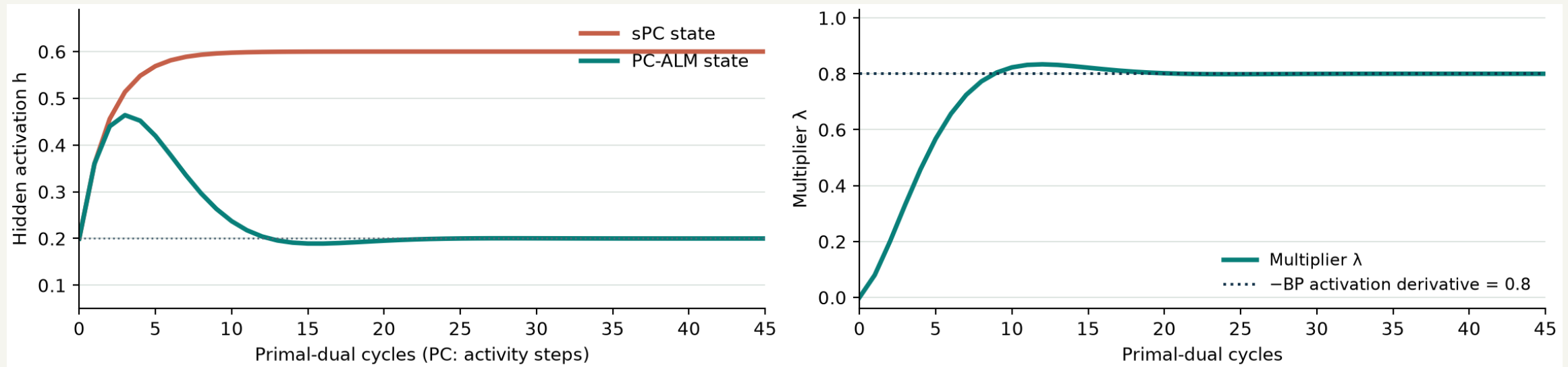
Convergence theorem: stable linear dynamics with fixed weights.

An endpoint characterization is not a finite-step or nonlinear convergence guarantee.

A scalar example makes the difference visible

Same weights and sample as before; only the inference rule changes.

$$x = y = 1 \quad w_1 = 0.2 \quad w_2 = 1 \quad \rho = 1 \quad \eta = 0.2 \quad \alpha = 0.5$$



Weight gradients (w_1, w_2)

sPC: $(-0.40, -0.24)$

ALM = BP: $(-0.80, -0.16)$

Independent teaching calculation. Traces show repeated cycles; endpoint values are checked.

The paper observes faster credit propagation

Local recurrence produces a traveling credit signal in deep networks.

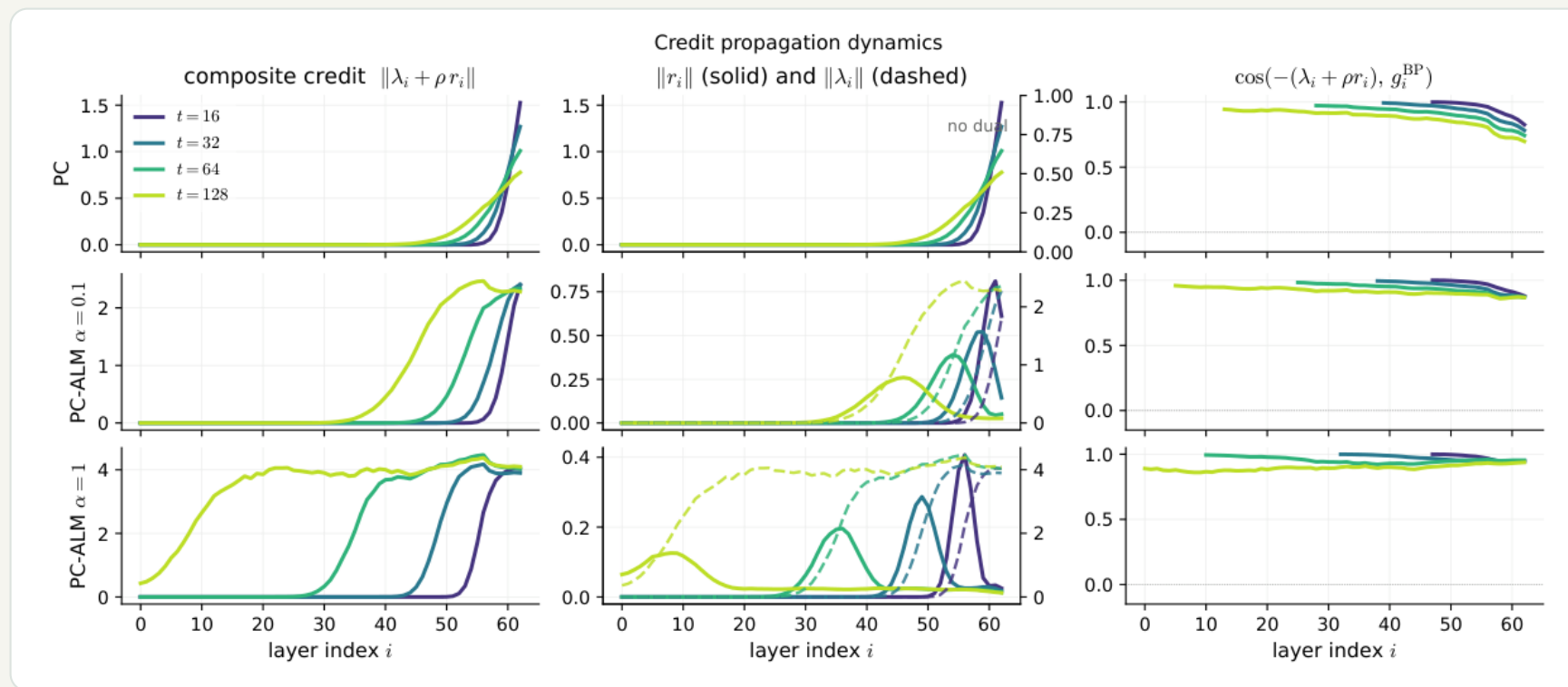


FIGURE 3

PC: credit near the output.

ALM: credit reaches earlier layers.

Width 32; depth 64.
One sample at initialization.

“Ballistic” describes a wavefront; it does not imply instant credit or complete convergence.

The headline result: the deep–narrow gap closes

Reported Fashion-MNIST results at a matched PC inference budget.

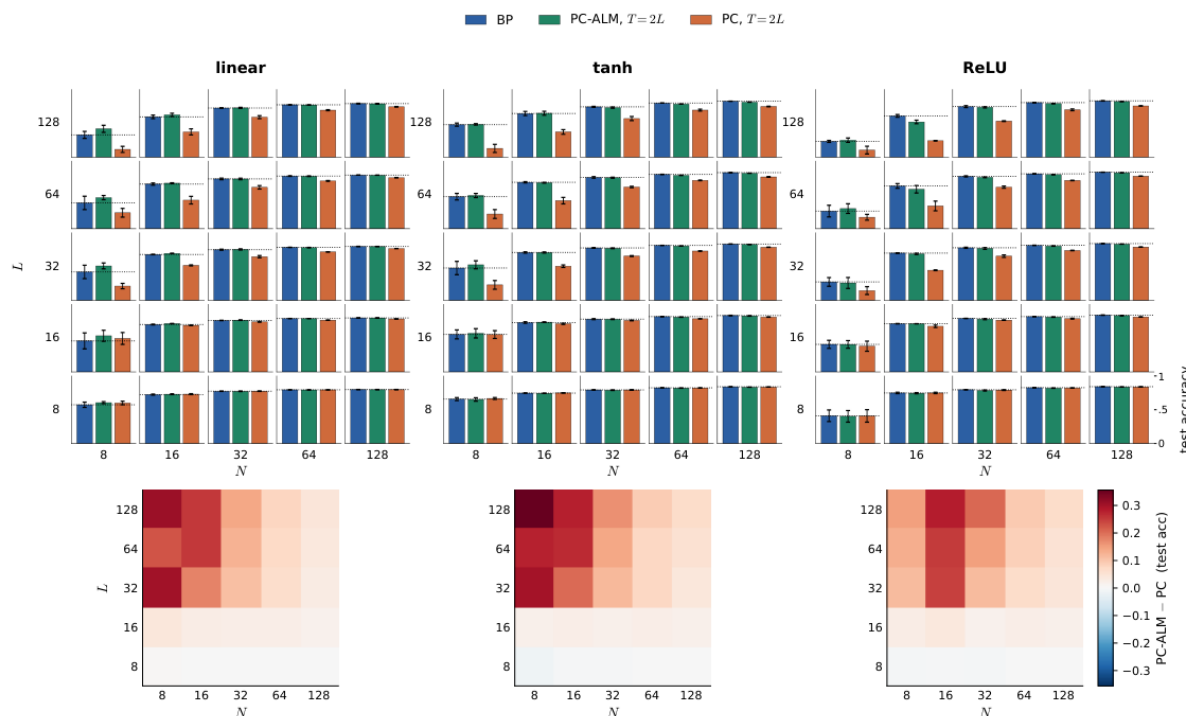


FIGURE 2 • THE TESTED GRID

Widths + depths
8, 16, 32, 64, 128

Identity, tanh, ReLU

$T = 2L$ activity steps

One training epoch

Reported BP-level performance in this grid. No ePC baseline or long-training result.

Promising mechanism; limited practical evidence

Separate endpoint theory, finite-budget accuracy, and measured cost.

| Theory

BP endpoint identity.

Conditional linear convergence.

Nonlinear convergence remains open.

| Evidence

Two small datasets.

Residual MLPs.

One epoch; no FabricPC or ePC benchmark.

| Cost

Extra dual tensor and residual work.

Coupled rate tuning.

Measure actual time and memory.

The paper's compute gain is versus widening PC for credit alignment, not versus BP or ePC.

Three mechanisms, three different tradeoffs

Assume a matched acyclic graph, loss, penalty, initialization, and scaling.

Property	sPC	ePC in FabricPC	PC-ALM
Relaxed variables	Hidden states h	Errors ϵ ; derive h	States h + multipliers λ
Inference coupling	Adjacent-layer derivatives	Full derived-graph reverse pass	Adjacent primal-dual updates
Target endpoint	Soft PC-energy stationary point	Same PC-energy stationary points*	BP at feasible stationary point
Main tradeoff	Simple local energy; finite-penalty gap	Fast signal access; global AD + curvature	Local BP credit; extra state + stability

* Compatible acyclic objective and gradients. Cyclic unrolling and nonlinear μ PC need separate treatment.

An optimized error ϵ and an accumulated multiplier λ are different variables with different jobs.

FabricPC needs a coordinated extension

Proposed design based on the inspected source; no solver has been implemented here.

| State contract

Per-example hidden dual state; initialization, reset, pytrees, batching.

| Objective + learning

Augmented local term in both activity and weight derivative paths.

| Inference lifecycle

New-state residuals; exact dual schedule; final primal-only step.

| Diagnostics + scope

Residual, dual, credit, stationarity; audit clamps, energies, scaling.

Start with an acyclic Gaussian model and extend the shared contracts consistently.

Evaluate correctness before model-scale claims

A proposed experiment sequence with explicit comparisons and decision criteria.

1 Algebra

Scalar + linear graphs.

$\alpha = 0$ reproduces sPC.

Settled stable ALM
gradients match BP.

2 Fair baselines

BP, sPC, both ePC regimes,
PC-ALM.

Measure gradient error,
time, memory, and
stability.

3 Transfer

Reproduce the paper.

Then longer training and
CIFAR conv models.

Multiple seeds + tuned
rates.

Success should be a measured benefit at comparable cost, with the learning objective stated.

Recommendation: test PC-ALM as a research method

The opportunity is local BP-consistent credit; the practical win must be measured.

Build a bounded research implementation and compare it with the ePC we actually use.

| If locality is the objective

PC-ALM is a compelling mechanism to test.

| If GPU speed is the objective

The advantage over ePC is still unmeasured.

A zero residual can coexist with a nonzero learning signal: the multiplier carries the credit.

Discussion: what should the first experiment prove?

Choose the primary objective before choosing the benchmark.

- 01** **Local dynamics that reproduce BP credit?**
- 02** **Better learning in deep, narrow networks?**
- 03** **Lower time or memory than our current ePC?**

Proposed first step: a linear acyclic solver, then one matched deep–narrow experiment.

Notation and conventions

Backup · one symbol, one meaning in this presentation

$\mathbf{h}_i / \mathbf{W}_i$	Hidden activation / layer weight matrix
$\mathbf{r}_i / \boldsymbol{\varepsilon}_i$	Measured residual / optimized ePC error coordinate
$\lambda_i / \mathbf{c}_i$	Multiplier / composite credit $\lambda_i + p\mathbf{r}_i$
$\boldsymbol{\delta}_i / \mathbf{J}_{i+1}$	BP activation derivative / adjacent layer Jacobian
ρ / α	Penalty strength / dual ascent rate
$\eta / \eta \boldsymbol{\varepsilon} / \eta \theta$	Activity / error-coordinate / weight step sizes
$\mathbf{L} / \mathbf{N} / \mathbf{T}$	Depth / width / inference activity-step count

Sign convention: $r = \text{state} - \text{prediction}$; at the ALM endpoint $\lambda = -\delta$.

Derive the scalar example by hand

Backup • all endpoint values follow from two scalar derivatives

$$h_{\text{PC}}^* = \frac{w_2 y + \rho w_1 x}{w_2^2 + \rho}$$

$$h_{\text{ALM}}^* = w_1 x, \quad \lambda^* = w_2 (y - w_2 w_1 x)$$

$$\nabla_{w_1} \mathcal{L}_\rho = -\lambda^* x, \quad \nabla_{w_2} \mathcal{L}_\rho = (w_2 w_1 x - y) w_1 x$$

At $x = y = w_2 = \rho = 1$ and $w_1 = 0.2$: $h_{\text{PC}} = 0.6$; $h_{\text{ALM}} = 0.2$; $\lambda_{\text{ALM}} = 0.8$.

Why ePC changes dynamics but preserves the objective

Backup • acyclic error coordinates induce a different state-space metric

$Q = \partial h / \partial \epsilon$ is invertible and block lower triangular on a DAG.

$$\nabla_{\epsilon} E = Q^{\top} \nabla_h F_{PC}$$

$$\Delta h \approx -\eta_{\epsilon} Q Q^{\top} \nabla_h F_{PC}$$

Same stationary points; changed geometry and trajectories.

At a stationary point: $H_{\epsilon} = Q^{\top} H_h Q$.

Global signal access comes from the error-to-state dependency graph, not a new objective.

Linear stability is a property of the joint update

Backup · the simple bound omits the supervised-loss curvature block

Linear residual $\mathbf{r} = \mathbf{A}\mathbf{h} + \mathbf{b}$; readout $\hat{\mathbf{y}} = \mathbf{C}\mathbf{h}$; curvature $\mathbf{B} = \mathbf{C}^\top \mathbf{C}$.

$$\mathbf{K} = \mathbf{I} - \eta_h(\mathbf{B} + \rho \mathbf{A}^\top \mathbf{A})$$

$$\mathbf{M} = \begin{pmatrix} \mathbf{K} & -\eta_h \mathbf{A}^\top \\ \alpha \mathbf{A} \mathbf{K} & \mathbf{I} - \alpha \eta_h \mathbf{A} \mathbf{A}^\top \end{pmatrix}$$

Require spectral radius $\text{spr}(\mathbf{M}) < 1$. If $\mathbf{B} = 0$ only: $\eta s^2(2\rho + \alpha) < 4$ for every singular value s of $\mathbf{A}$.

The constraint-only bound is not a universal guarantee for a nonlinear supervised model.

Integration details that affect the algorithm

Backup · objective semantics and update order are part of correctness

■ Preserve mathematical meaning

error remains state – prediction.

Duals belong to examples.

The supervised loss is not a hidden equality constraint.

■ Audit all derivative paths

Recomputed residual after primal.

Augmented term in local learning.

Scaling, custom energies, cycles, and masks require explicit scope.

A buffer containing composite credit cannot substitute for the augmented objective contract.

Sources and reproducibility

Backup • paper claims, repository observations, and proposals are kept distinct

[P] Seely & Gould • Augmented Lagrangian Predictive Coding

arXiv:2605.31022v1 • supplied paper • Algorithm 1; Appendices A, C–G

[E] Goemaere et al. • ePC: Fast and Deep Predictive Coding in Digital Simulation

arXiv:2505.20137 • implementation comparison grounded in FabricPC

FabricPC source snapshot • 22 September 2026

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[I] inference [R] ePC [N] state types [W] learning [G] guide [D] report [T] tests

Full links and derivations are in the private speaker guide. Code and plots are reproducible.